

Course: Civil Engineering
Subject Code: CENUGPC18

Year: Third Year
Max. Marks: 80

Subject: Environmental Engineering
Time: 3 hrs

1. Answer any two questions: (15X2=30)

a) Enumerate the factors on which per capita demand depends? (5)

Following is the population of a city as noted from the census department:

Year	Population
1950	35000
1960	78000
1970	115000

Determine a) the saturation population, b) expected population in 1980 and the equation of the logistic curve? (10)

b) Water has to be supplied to a town of population 100000. If the water works is situated at a lower elevation of 100 m than the water level in the source, then determine the size of the gravity main which is required to convey the water from the source to the water work. Consider the length of the gravity main as 40 km and the per capita demand of the town is 130 l/day/capita. (15) $f = 0.013 - 0.015$, 8-12 hrs pumping

c) A main combined sewer was designed to serve an area of 150 sq. km with an average population of 200 persons/hectare. The average rate of sewage flow is 450 litres/capita/day. The maximum flow is 60% in excess of the average together with the rainfall equivalent of 15 mm in 24 hours, all of which are run off. What should be the capacity of the sewer in cu.m/sec. (15)

2. Answer any two questions: (25X2=50)

a) i) In a continuous flow settling tank 4.5 m deep and 80 m long, what flow velocity of water would you recommend for effective removal of 0.025 mm particles at 25° C. The sp. gravity of particles is 2.65, and kinematic viscosity ν for water may be taken as 0.01 cm²/sec. (8) $T = 1 \text{ cm/s}$

ii) A main combined sewer was designed to serve an area of 60 sq. km with an average population of 185 persons/hectare. The average rate of sewage flow is 350 litres/capita/day. The maximum flow is 50% in excess of the average together with the rainfall equivalent of 12 mm in 24 hours, all of which are run off. What should be the capacity of the sewer in cu.m/sec (8)

iii) Find the minimum velocity and gradient required to transport coarse sand through a sewer of 40 cm diameter with sand particles of 1mm dia and specific gravity 2.65. Assume k for sand = 0.04. The Manning's roughness coefficient (n) for the sewer material may be assumed as 0.012. (9)

b) A 80 cm diameter well is being pumped at a rate of 1,560 litres/minute. Measurement in a nearby test well were made at the same time as follows. At a distance of 6 m from the well being pumped, the drawdown was 7.2 m, and at 15 m, the drawdown was 2.7 m. The bottom of the well is 100 m below the ground water table. (a) Find out the coefficient of permeability. (b) If all the observed pts. were on the Dupuit curve, what was the drawdown in the well during pumping? (c) What is the sp. capacity of the well? (d) What is the maximum rate at which water can be drawn from this well? (8+8+9=25)

c) i) Differentiate between slow sand and rapid sand filter. (7)

ii) What is meant by plain sedimentation? Explain breakpoint chlorination. Derive an expression for settling velocity of particles. (2+3+3=8)

iii) Describe how the storage capacity of a reservoir can be determined graphically as well as analytically. (10)

6TH SEMESTER B. TECH (CIVIL ENGINEERING)

END SEMESTER EXAMINATION, MAY-2022

MATERIAL SCIENCE & METALLURGY
MENUGOE01

Full Marks = 80

Time – 3 hours

(Answer any FOUR)

- ✓ 1. (a) Classify Engineering materials. Briefly describe their properties and applications. 8
- ✓ (b) What are the criteria of selection of materials? Explain 5
- ✓ (c) What are the 14 Bravais Lattice? 7
- ✓ 2. (a) Show that the Atomic Packing Fraction for HCP crystal is 0.74 12
- ✓ (b) What are the Miller indices of a plane intersecting the x-axis at 2, y-axis at $\frac{1}{2}$, and parallel to z-axis? The structure is a cube. Sketch. 8
- ✓ 3. (a) Determine the size of a tetrahedral void in a unit cell. 8
- (b) Show $(10\bar{1})$; (010) planes, and $[11\bar{1}]$; $[\bar{1}00]$ directions in a cubic unit cell 4
- (c) Briefly explain the different types of crystal imperfections. 8
- ✓ 4. (a) If a first order X-ray, having wavelength of $1.72\text{\AA}$, showing diffraction angle of 30° using (111) plane of a BCC crystal, Find the radius of one atom of the material forming the crystal. 10

[TURN OVER]

(2)

- ✓ (b) Copper (63.5) has a density of 5500 kg/m^3 . It has FCC structure. Find out the radius of one atom of Copper. 10
- ✓ 5. (a) Deduce the expression for fraction of vacancy site ($\frac{N_v}{N}$). 12
- ✓ (b) The fraction of vacancy site ($\frac{N_v}{N}$) in a crystal is 1×10^{-10} at 500°C . Determine the fraction at 1000°C . 8
6. (a) Explain Edge and Screw Dislocations with neat sketches. 12
- (b) What are the two types of dislocation motions? Compare them. 8

ALIAH UNIVERSITY
End-Term Examination- 6th Semester
Session-2021-2022

Year: 3rd Year
Semester: 6th Semester
Subject: Foundation Engineering
Subject Code: CENUGPC13

Full Marks: 80
Time: 3hr.

Answer question no.1 and any three from the rest:

1. Answer the question briefly

[4*5=20]

- ☒ i) What is shallow foundation? And state the types of shallow foundation.
 - ☒ ii) Explain the different modes of failure of foundation soil.
 - ☒ iii) Briefly state the disadvantages of direct shear test.
 - ☒ iv) Briefly discuss the properties and uses of floe net.
 - ☒ v) What is Terzaghi's one dimensional consolidation theory? Write the assumptions made in his theory?
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- ☒ 2. a) What is the need to know the maximum pre-consolidation stress of a soil stratum? Discuss an empirical method to determine pre-consolidation stress. [10]
 - ☒ b) A clay soil, tested in a consolidometer, showed a decrease in void ratio from 1.20 to 1.10 when the pressure was increased from 0.25 to 0.50 kgf/cm². Calculate coefficient of compressibility (a_v) and coefficient of volume compressibility (m_v). If the coefficient of consolidation (c_v) determined in the test for the given stress increment was 10 m²/year, calculate the coefficient of permeability in cm/s. If the sample tested at the site was taken from a clay layer 3.0 m in thickness, determine the consolidation settlement resulting from the given stress increment. [10]
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- ☒ 3. a) i) Write Coulomb's shear strength equation. ii) State the different method of measurements of shear strength parameters. iii) What are the different types of triaxial tests? [2+3+5]
 - ☒ b) Derive with illustration the torque required in field vane shear test. [10]
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- ☒ 4. a) i) What are the factors which influence bearing capacity of a shallow foundation? ii) What is unconfined compressive strength of a soil and how it is related with the shear parameter? [4+6]
 - ☒ b) i) What is bearing capacity? ii) Define angle of repose.
 - ☒ iii) Determine the bearing capacity of a strip footing, 1.5 m wide, with its base at a depth of 1 m, resting on a dry sand stratum with: Use Terzaghi's theory.

$$\gamma_s = 17 \text{ kN/m}^3, \phi' = 38^\circ \text{ and } c' = 0.$$

[1+2+7]

5. a) What do you understand by UU test, CU test and CD test. Discuss the time required by each of the tests.

[8]

b) In a direct shear test on a sand sample, the normal stress was 200 kN/m^2 and the sample failed at a shear stress of 120 kN/m^2 . Draw the Mohr circle and the strength envelope. Determine the (i) angle of shearing resistance of the soil (ii) the magnitude of the major and minor principle stresses, and (iii) the orientation of the principle planes with respect to horizontal.

[12]

6. i) Classify piles based on different criteria. ii) Define end bearing pile and friction pile.

[4+2+2]

i) Define negative skin friction.

ii) Discuss different types of vertical stress distribution diagram with neat sketches as per Boussinesq equation.

[2+10]

Aliah University

End Semester Examination – 2021-22

Course: Civil Engineering; Year: 3rd Year; Subject: Railway Engineering;

Subject Code: CENUGPE01(PEC-I); Max. Marks: 80; Times: 3 hrs.

Answer all questions.

Q.1 a) Write short notes on the following.
(2X5=10)

(i) Organizational Structure of Indian Railways 14

(ii) Minimum depth of ballast cushion. 18

(b) Define sleeper density. List the factors which effects the sleeper density. For a sleeper density of M+7, calculate the number of sleepers in one km of rail length.

(2+4+4)

Q. 2. (a) What are the various resistances to the movement of train on the track? Discuss.

(10)

(b) Discuss the problems associated with multigauge system. 69

(10)

Q.3. (a) Discuss the requirements of an ideal rail joint. 126

(10)

(b) Describe the different measures to reduce the creep in rails. 140

(10)

Q.4 What do you mean by railway alignment? What are the requirements of an alignment? Discuss the factors controlling the alignment. Also discuss the necessity for geometric design of alignment. 127 213

(2+5+7+6)

ALIAH UNIVERSITY
End Semester Examination-2021-22

B. Tech (6th Semester)

Department of Civil Engineering

Subject Code: CENUGPE02

Subject Name: Engineering Hydrology

Time: 3 Hr

Maximum Marks= 80

Note: Attempt any 5 questions. All questions carry equal marks.

1. Describe the hydrological cycle. List the major activities in which hydrological studies are important.

2. A catchment has six rain gauge stations. In a year, the annual rainfall recorded by the gauge are as follows:

station	A	B	C	D	E	F
rainfall (cm)	82.6	102.9	180.3	110.3	98.8	136.7

(a) Determine the standard error in estimation of mean rainfall in the existing set of rain gauge

(b) For a 10% error in the estimation, calculate the optimum number of rain gauge.

3. A storm with 10 cm of precipitation produced a direct runoff of 5.8 cm. The duration of the rainfall was 16 hours and its time distribution is given below.

Time from start (h)	0	2	4	6	8	10	12	14	16
Cumulative rainfall (cm)	0	0.4	1.3	2.8	5.1	6.9	8.5	9.5	10.0

Estimate ϕ index of the storm.

4. Discuss the factor affecting runoff hydrograph and components of hydrograph.

5. A storm over a catchment of area 5.0 km² had duration of 14 hours. The mass curve of rainfall of the storm is as follows:

Time from start (h)	0	2	4	6	8	10	12	14
Cumulative rainfall (cm)	0	0.6	2.8	5.2	6.6	7.5	9.2	9.6

If the ϕ index for the catchment is 0.4 cm/h, determine the effective rainfall hystograph and volume of direct runoff from the catchment due to the storm.

6. Two storms, each of 6-hr duration and having rainfall excess values of 3.0 cm and 2.0 cm respectively, occur successively. The 2 cm ER rain follows the 3 cm rain. The 6 h unit hydrograph for the catchment is as :

Time (h)	0	3	6	9	12	15	18	24	30	36	42	48	54	60	69
UH ordinate (m ³ /s)	0	25	50	85	125	160	185	160	110	60	36	25	16	8	0

Calculate the resulting DRH.